

A dense sequence with distinct products of consecutive blocks

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July 2, 2026

Abstract

We construct a set $A \subset \mathbb{N}$ of natural density one whose increasing enumeration $d_1 < d_2 < \dots$ has the property that the products over finite consecutive blocks are all distinct. The set is produced by a greedy algorithm, modified by a deterministic deletion set of density zero. The role of this deletion set is to convert the semiprimes supplied by the Matomäki–Teräväinen theorem into private barriers: short obstructions carrying a globally unique large prime. Once a few density-zero exceptional families are discarded, the barriers confine each surviving collision to a polylogarithmic window, and the resulting product identities carry many large-prime congruences of shifted-CRT type. A low-degree auxiliary polynomial interpolating the local data is then forced to vanish at the global collision point. Its nonlinear components are counted by the Castryck–Cluckers–Dittmann–Nguyen affine-curve bound, and its line components directly from the curvature of q^θ . The outcome is $o(X)$ surviving collision endpoints in each dyadic shell $(X, 2X]$, and hence a density-one set with distinct consecutive-block products.

2020 Mathematics Subject Classification. Primary 11B75; Secondary 11N25, 11N35, 11D45.

Key words and phrases. Distinct products, consecutive blocks, natural density, greedy construction, smooth numbers, determinant method, Erdős problems.

1 Introduction

An increasing sequence of positive integers assigns to each finite interval of indices the product of the corresponding terms, and one may ask how rigidly those block products pin down the interval. For an increasing sequence

$d_1 < d_2 < \dots$, write

$$P_d(u, v) = \prod_{i=u}^v d_i \quad (1 \leq u \leq v).$$

Say that the sequence has *distinct consecutive-block products* if

$$P_d(u, v) = P_d(u', v') \implies (u, v) = (u', v')$$

for all finite index intervals, so that no two different blocks share a product. Such rigidity is easy to arrange in a thin sequence; the question is how dense one can make it. Erdős asked whether it is compatible with natural density one. This is Problem 421 in the Erdős problem collection [2], going back to the problem list of Erdős and Graham [1].

Theorem 1.1. *There exists a strictly increasing sequence*

$$1 \leq d_1 < d_2 < \dots$$

whose underlying set has natural density 1, and for which the products $\prod_{i=u}^v d_i$ are all distinct as (u, v) ranges over finite consecutive intervals.

Our argument is self-contained except for the external results collected in Section 2. The sequence is built greedily, with one modification. We fix in advance a deterministic set D of density zero, reject every integer of D outright, accept 2, and accept each later integer outside D precisely when doing so creates no new coincidence among consecutive-block products. Uniqueness is then automatic from the greedy rule, and the whole difficulty lies in showing that the rejected integers – those forced by D together with those the greedy test discards – still form a set of density zero.

To locate that difficulty, suppose an integer $n \notin D$ is rejected. Then adjoining n produces a collision, and a short analysis (Lemma 3.3) shows that the collision is always an *old-new witness*

$$P = nS,$$

in which P is a block product already present and S is a product of the most recent accepted terms, possibly empty. When $S = 1$, the integer n is itself an old block product, and such n are sparse for elementary reasons. The genuine obstruction is the case $S > 1$: here P may be long and may sit far to the left of the suffix, so no direct count is available. The set D is designed precisely to force almost every such witness into a short window, where it can be counted.

This window is produced by what we call a *barrier system*. In most blocks of polylogarithmic length we single out one semiprime $b = p_1 p_2$ with $(\log b)^{1.08} < p_1 \leq (\log b)^{1.1}$ whose large prime factor p_2 is globally unique among our choices, and we delete every other multiple of p_2 up to a long but finite horizon.

Two properties make this effective. A selected barrier survives the greedy scan, because any new product through it inherits the private prime p_2 that no earlier product can carry; hence the barrier is accepted. And once a barrier is in place, no surviving collision can have its old block cross it, unless the endpoint has already passed the barrier's horizon. The integers beyond their horizon make up a smooth family of density zero, removed separately.

With the barriers in place, every nonempty collision left in a shell $(X, 2X]$ has a short, tightly controlled witness

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j), \quad 0 \leq a_i, c_j \leq (\log X)^{O(1)},$$

with $x < q < n \leq 2X$, q a previously selected barrier, and

$$x = q^{b/\ell} + O((\log X)^B), \quad (\log X)^{-B} \leq 1 - \frac{b}{\ell} \ll \frac{\log \log X}{\log X}.$$

All the shifts a_i, c_j are now bounded by a fixed power of $\log X$, and the argument becomes arithmetic. A large prime dividing a factor on the left must reappear among the terminal factors on the right, and matching these yields many congruences

$$p_i \mid x - a_i, \quad p_i \mid q - x + a_i + c_i,$$

with distinct $p_i > \exp(\log X / (\log \log X)^2)$. A low-degree auxiliary polynomial that vanishes on the local points $(a_i, a_i + c_i)$ is therefore forced to vanish at the global point $(x, x - q)$ as well. Its nonlinear components are counted by the affine-curve determinant-method theorem of Castryck–Cluckers–Dittmann–Nguyen, while its line components are handled directly, through the strict curvature of q^θ for $\theta < 1$.

The paper is organized as follows. Section 2 fixes the external inputs and notation. Section 3 records the formal greedy reduction and disposes of empty-suffix collisions. Section 4 constructs the barrier system. Sections 5 and 6 turn the barrier information into localized product identities and then into terminal large-prime support. Section 7 proves the shifted-CRT counting proposition. Section 8 applies it to all remaining nonempty rejections, and Section 9 completes the density argument.

2 External Inputs and Notation

All logarithms are natural, except $\log_2$. Constants implicit in $O(\cdot)$, $o(\cdot)$, and $\ll$ may depend on the fixed constants chosen in the proof, but not on the dyadic parameter X . A *dyadic shell* means $(X, 2X]$, with X a sufficiently large power of two. For an integer $m \geq 2$, let $P^+(m)$ be its largest prime factor. Let

$$\Psi(X, Y) = \#\{m \leq X : P^+(m) \leq Y\}.$$

External input 2.1 (Matomäki–Teräväinen E_2 theorem [4]). *There are constants $c_0, \delta > 0$ such that, for all sufficiently large X , all but $O(X/(\log X)^\delta)$ integer starts $2 \leq y \leq X$ satisfy*

$$\#\left\{p_1 p_2 \in (y, y + (\log y)^{2.1}] : \begin{array}{l} p_1, p_2 \text{ prime,} \\ (\log y)^{1.09} < p_1 \leq (\log y)^{1.1} \end{array} \right\} \geq c_0 (\log y)^{1.1}.$$

External input 2.2 (de Bruijn–Dickman smooth-number estimate [5, 6]). *Uniformly for $u = u(N) \rightarrow \infty$ with $u = O(\sqrt{\log \log N})$,*

$$\Psi(N, N^{1/u}) \leq N \exp\left(-(1 + o(1))u \log u\right).$$

External input 2.3 (CCDN affine-curve bound [7]). *For every fixed ambient dimension $n > 1$, there is a constant C_n such that, for every $B \geq 1$, every integral affine curve $V \subset \mathbb{A}_{\mathbb{Q}}^n$ of positive degree d satisfies*

$$\#\{z \in V(\mathbb{Z}) : \text{ht}(z) \leq B\} \leq C_n d^3 B^{1/d} (\log B + d).$$

Here $\text{ht}(z) = \max_i |z_i|$ for $z = (z_1, \dots, z_n) \in \mathbb{Z}^n$. The constant C_n is independent of d , B , and the defining equations of V ; the displayed factor is the only degree-dependence used below.

External input 2.4 (Mignotte factor inequality [8]). *If $f, g \in \mathbb{Z}[z]$ are nonzero, $\deg g \leq d$, and g is a multiple of f , then*

$$\|g\|_2 \geq 2^{-d} \|f\|_\infty.$$

We also use the following standard elementary facts: Bertrand’s postulate, Chebyshev upper and lower prime-counting bounds, Gauss’s lemma for primitive factors over $\mathbb{Q}$ and $\mathbb{Z}$, the usual compatibility of primitive homogenization and dehomogenization for line factors, Galois conjugacy of absolute factors of a squarefree $\mathbb{Q}$ -irreducible plane curve, and Bezout’s $d_1 d_2$ bound for two coprime plane curves of degrees d_1, d_2 .

3 The Modified Greedy Reduction

It is convenient to isolate the purely formal consequences of the greedy rule from the number-theoretic construction of D , which is deferred to Section 4. Granting that D has density zero, the only rejections that will demand real work are the nonempty old-new collisions introduced below.

Definition 3.1 (Modified greedy sequence). Fix a deterministic set $D \subset \mathbb{N}$ with $1 \in D$, $2 \notin D$, and $|D \cap [1, N]| = o(N)$. The modified greedy construction rejects every $n \in D$, accepts 2, and for each $n \geq 3$ with $n \notin D$, accepts n if and only if adjoining n to the accepted prefix creates no equality between two distinct finite consecutive-block products.

Lemma 3.2 (Greedy uniqueness). *Every finite accepted prefix produced by Definition 3.1 has distinct consecutive-block products.*

Proof. The initial accepted prefix is $\{2\}$. A rejection leaves the accepted prefix unchanged. An acceptance occurs only after the enlarged prefix has been tested to have no equality between two distinct consecutive-block products. Induction over the scan proves the claim. $\square$

Lemma 3.3 (Old-new witness split). *Let $n \geq 3$ be rejected by the modified greedy construction and suppose $n \notin D$. If $d_1 < \dots < d_m$ is the accepted prefix before testing n , then there is an old consecutive accepted block product P and a suffix product S of the old prefix, possibly $S = 1$, such that*

$$P = nS.$$

Proof. The old prefix has distinct block products by Lemma 3.2. Since $n \notin D$, rejection means that adding n creates a collision. Old-old collisions are impossible. Two new block products both contain n ; after cancelling n , one obtains equality of two final suffix products of the old prefix. Distinct final suffixes are strictly ordered by inclusion, and extending a suffix multiplies its product by an accepted integer at least 2. Hence two new products cannot collide. The collision is therefore between an old block and a new block, and every new block is n times an old suffix product. $\square$

Definition 3.4. A nonforced rejection $n \notin D$ is an *empty-suffix rejection* if it has some witness in Lemma 3.3 with $S = 1$. Otherwise we choose, once and for all, the lexicographically first old-new witness in accepted-index coordinates, first by old-block endpoints and then by suffix start, and call n a *nonempty rejection*. For nonempty rejections we cancel the maximal common

contiguous overlap between the old block and the final suffix. The old block is not contained in that suffix, and if the cancellation exhausted the suffix then n would itself be an old block product. Thus the remaining suffix is nonempty, and the old block lies strictly before it.

Lemma 3.5 (Sparse old products). *For any increasing positive-integer sequence $d_1 < d_2 < \dots$ with $d_1 \geq 2$, the number of values at most N which occur as $P_d(u, v)$ with $v - u + 1 \geq 2$ is $o(N)$.*

Proof. Since $d_i \geq i + 1$, a length- ℓ block starting at index i has product at least $(i + 1)^\ell$. For fixed $\ell \geq 2$, at most $N^{1/\ell}$ length- ℓ block products can be $\leq N$. No length $\ell > \log_2 N$ contributes. Thus the number of such values is at most

$$\sum_{2 \leq \ell \leq \log_2 N} N^{1/\ell} = O(N^{1/2} \log N) = o(N).$$

□

Corollary 3.6. *The empty-suffix rejections of the modified greedy construction have density zero.*

Proof. An empty-suffix rejection n is an old block product. The old block has length at least two, since length one would make n an already accepted old term. Apply Lemma 3.5 to the accepted sequence. □

Lemma 3.7 (Elementary smooth decay). *Let $E > 0$, $\rho > 1$, and $M > 0$ be fixed. Then*

$$\Psi(X, (\log X)^E) = o(X)$$

and, with $Y_X = \exp(\log X / (\log \log X)^\rho)$,

$$\Psi(2X, Y_X) = o\left(\frac{X}{(\log X)^M}\right).$$

Proof. For $1/2 \leq \sigma < 1$, Rankin's trick gives

$$\Psi(X, Y) \leq X^\sigma \prod_{p \leq Y} (1 - p^{-\sigma})^{-1}.$$

Write $\sigma = 1 - \eta$, $0 < \eta \leq 1/2$. Chebyshev's bound and partial summation give

$$\log \prod_{p \leq Y} (1 - p^{-\sigma})^{-1} \ll \log \log(3Y) + \frac{Y^\eta}{\eta \log Y}.$$

This rests on the estimate

$$\sum_{p \leq Y} p^{-1+\eta} \leq \pi(Y)Y^{-1+\eta} + (1-\eta) \int_2^Y \pi(t)t^{-2+\eta} dt \ll \frac{Y^\eta}{\log Y} + \int_2^Y \frac{t^{-1+\eta}}{\log t} dt,$$

and splitting the last integral at $\log t = 1/\eta$, which gives $O(\log \log(3Y))$ below the split and $O(Y^\eta/(\eta \log Y))$ above it.

For $Y = (\log X)^E$, choose fixed $\sigma \in (0, 1)$ with $E(1-\sigma) < 1$, obtaining $X^\sigma \exp(o(\log X)) = o(X)$. For $Y_X = \exp(L/u)$, $L = \log X$, $u = (\log \log X)^\rho$, take $\eta = (\log u)/(2 \log Y_X)$. Then $Y_X^\eta = \sqrt{u}$ and $\eta L = \frac{1}{2}u \log u$, while the Euler-product exponent above is

$$O(\log \log X + \sqrt{u}/\log u) = o(u \log u).$$

Therefore

$$\Psi(2X, Y_X) \leq 2X \exp(-\Omega(u \log u)),$$

which beats every fixed power of $\log X$ because $\rho > 1$. $\square$

Lemma 3.8 (Actual blocks and final suffixes). *For any strictly increasing accepted set:*

- (a) *for fixed accepted endpoint x and fixed length ℓ , there is at most one actual consecutive accepted block of length ℓ ending at x ;*
- (b) *for fixed $r \geq 0$, the function*

$$F_r(n) = n \prod_{\substack{a < n \\ a \text{ accepted} \\ a \text{ among last } r}} a$$

is strictly increasing on candidates n for which the last r accepted predecessors exist.

Proof. Part (a) is immediate from strict increase. If $n_1 < n_2$, then the ordered r -tuple of last accepted predecessors before n_1 is coordinatewise $\leq$ the corresponding tuple before n_2 . The predecessor product is therefore nondecreasing, and multiplication by $n_1 < n_2$ gives $F_r(n_1) < F_r(n_2)$. $\square$

4 Private Barriers

We turn to the construction of the deterministic set D . The barriers are built only here; everything afterwards rests solely on the three properties recorded in Proposition 4.1. Put

$$\beta = 6, \quad L_X = \left\lfloor (\log X)^\beta \right\rfloor, \quad L_0(X) = (\log(3X))^{2.1}.$$

For each sufficiently large dyadic shell $(X, 2X]$, partition the shell into full consecutive intervals of length L_X ; boundary remainders are put in D . In a full block $I = [U, U + L_X)$, an integer start y is *admissible* if $U \leq y$ and $y + L_0(X) < U + L_X$.

Proposition 4.1 (Finite-horizon private barriers). *There is a deterministic set $D \subset \mathbb{N}$ of density zero and a set of selected semiprime barriers $b = p_1 p_2$ such that:*

(i) *every selected barrier has*

$$(\log b)^{1.08} < p_1 \leq (\log b)^{1.1},$$

and its large prime p_2 is a globally unique private label;

(ii) *every selected barrier lies outside D and is accepted by the modified greedy construction;*

(iii) *for every sufficiently large N , put*

$$F_Y = Y \lfloor \sqrt{\log \log Y} \rfloor,$$

and let $X_0 = X_0(N)$ be the dyadic scale with $F_{X_0} < N \leq F_{2X_0}$. Outside D , a finite initial segment, and the $o(N)$ inverse-horizon smooth family whose chosen witness has old-left maximum $x \leq 5X_0$, every nonforced nonempty rejection $n \leq N$, with its chosen overlap-cancelled witness, has a previous selected barrier $q_B(n) < n$ such that

$$n - q_B(n) = O((\log n)^\beta),$$

the final suffix of the witness starts after $q_B(n)$, and the old-left block in the witness has value span $O((\log n)^\beta)$.

Only the properties listed in Proposition 4.1 are used in the later localization and counting arguments.

Proof. We start above a large fixed dyadic shell, put $1 \in D$, keep $2 \notin D$, and absorb all finite initial choices into $O(1)$.

Good blocks and label supply. Call a full L_X -block bad if fewer than half of its admissible starts are good in the sense of Input 2.1. Since the admissible-start sets of distinct full blocks are disjoint and each full block has $(1 + o(1))L_X$ admissible starts, Input 2.1, applied up to $2X + O(L_X)$, implies that bad blocks in $(X, 2X]$ have total length $O(X/(\log X)^\delta)$. Put all bad blocks in D . For every admissible start in the shell, $y < 3X$ for large X , so $(\log y)^{2.1} \leq L_0(X)$; hence the MT interval attached to y remains inside its long block.

In any remaining block, the good starts contribute $\gg L_X(\log X)^{1.1}$ start-semiprime incidences. A fixed semiprime b can be counted only for starts y with $b - L_0(X) < y < b$, hence for at most $L_0(X) + O(1)$ starts. Thus the block contains at least

$$\gg \frac{L_X(\log X)^{1.1}}{L_0(X)} = (\log X)^{\beta-1-o(1)}$$

distinct semiprimes. A fixed large prime label p_2 gives at most one such semiprime in one long block: two products $p_1 p_2, p'_1 p_2$ differ by at least p_2 , while for these semiprimes $p_2 = b/p_1 \gg X/(\log X)^{1.1} \gg L_X$. Hence each good block has $(\log X)^{\beta-1-o(1)}$ distinct candidate labels. For every counted $b = p_1 p_2$, since $b \in (y, y + L_0(X)]$, we have $\log b = \log y + O(L_0(X)/X) = (1 + o(1)) \log X$. The upper bound below is immediate, and the lower bound follows from the exponent gap $1.09 > 1.08$; after increasing the initial cutoff,

$$(\log b)^{1.08} < p_1 \leq (\log b)^{1.1}.$$

Greedy block-label matching. Order good blocks by left endpoint and candidate labels in each block by increasing p_2 . Scan blocks in order and match a block to the first candidate label which was not selected in an earlier shell and has not already been matched in the current shell. If no such label exists, leave the block unmatched.

A label selected in an earlier shell $(Y, 2Y]$ can also be a candidate in $(X, 2X]$ only for polylogarithmically comparable Y . Comparing the two size ranges

$$\frac{X}{(\log X)^{1.1}} \ll p_2 \ll \frac{X}{(\log X)^{1.08}}, \quad \frac{Y}{(\log Y)^{1.1}} \ll p_2 \ll \frac{Y}{(\log Y)^{1.08}},$$

first gives $Y = X^{1+o(1)}$, and then the safer window

$$cX(\log X)^{-0.05} \leq Y \leq CX(\log X)^{0.05}.$$

Thus the number of earlier selected labels relevant to the current shell is

$$O\left(\frac{X}{L_X}(\log X)^{0.06}\right),$$

where the factor $(\log X)^{0.01}$ absorbs the $O(\log \log X)$ dyadic shells in the comparable range. Current-shell matched labels are absorbed in the same bound. Also a fixed label is adjacent to at most $(\log X)^{1.1+o(1)}$ current blocks: p_1 has only that many possible prime values, and each product $p_1 p_2$ lies in $O(1)$ long blocks. If U good blocks remain unmatched, then every candidate label of each unmatched block is either an earlier selected label or a label already matched in the current shell. Counting these incidences gives

$$U(\log X)^{\beta-1-o(1)} \ll \frac{X}{L_X}(\log X)^{0.06}(\log X)^{1.1+o(1)}.$$

Hence

$$U = O\left(\frac{X}{L_X}(\log X)^{-(\beta-2.16)+o(1)}\right) = o(X/L_X).$$

Put all unmatched good blocks in D . In each matched block select the matched semiprime as its barrier.

Prefix deletions. After every maximal deleted run of long blocks, and after each shell boundary, look at the first subsequent matched block and delete its nonbarrier prefix before the selected barrier. The number of bad blocks is $O(X/(L_X(\log X)^\delta))$, and the number of unmatched blocks is

$$O\left(\frac{X}{L_X}(\log X)^{-(\beta-2.16)+o(1)}\right).$$

Multiplying by the prefix length L_X , the block and prefix deletions in $(X, 2X]$ have size

$$O(X/(\log X)^\delta) + O\left(X(\log X)^{-(\beta-2.16)+o(1)}\right) + O(L_X) = o(X).$$

Finite-horizon multiples. For a selected barrier in shell X , let

$$C_X = \left\lfloor \sqrt{\log \log X} \right\rfloor, \quad F_X = X^{C_X}.$$

For its private label p_2 , put into D every multiple $m \neq b$ of p_2 with $m \leq F_X$. These horizon deletions have density zero. A shell $(X, 2X]$ contributes at most

$$O\left(\frac{\min(N, F_X)}{(\log X)^{\beta-1.1}}\right)$$

to $D \cap [1, N]$, since there are $O(X/L_X)$ selected labels and each label is $\gg X/(\log X)^{1.1}$. Put $s = \beta - 1.1 > 1$.

For old shells with $F_X < N$, the dyadic sum is dominated by the last such shell because C_X is nondecreasing and

$$F_{2X}/F_X \geq 2^{\frac{1}{2}\sqrt{\log \log X}}$$

eventually. This gives $O(N/(\log X_0)^s)$, where X_0 is the largest dyadic shell with $F_{X_0} < N$. Recent shells $X_0 < X \leq N$ contribute

$$N \sum_{X_0 < X \leq N} \frac{1}{(\log X)^s} = O\left(\frac{N}{(\log X_0)^{s-1}}\right) = o(N),$$

because $F_{X_0} < N \leq F_{2X_0}$ implies $\log X_0 \geq (1 + o(1)) \log N / \sqrt{\log \log N}$. Future shells $X > N$ can affect $[1, N]$ only if $X/(\log X)^{1.1} \leq N$, hence $X \leq N(\log N)^2$ for large N ; their total contribution is

$$O\left(\frac{N \log \log N}{(\log N)^s}\right) = o(N).$$

Finally, the other deterministic deletions have dyadic total $o(N)$. The boundary remainders contribute $O((\log N)^{\beta+1})$, and the MT-bad blocks contribute $O(N/(\log N)^\delta)$. The unmatched-block and prefix contribution is $O(X(\log X)^{-(\beta-2.16)+o(1)})$ in a shell; since $\beta = 6$, its dyadic total is also $o(N)$. This proves $|D \cap [1, N]| = o(N)$.

Barrier survival and acceptance. A selected barrier is in a matched block, is not in a deleted prefix, and is protected from its own label deletion. Nor can any other selected label delete it: if a selected label p from shell Y divides $b = p_1 p_2$ and $b \leq F_Y$, then $p = p_1$ or $p = p_2$. The case $p = p_2$ contradicts global label uniqueness. The case $p = p_1$ is impossible for large Y , because

$$p \geq \frac{Y}{(\log Y)^{1.1}}, \quad p_1 \leq (\log b)^{1.1} \leq (\log F_Y)^{1.1} = (C_Y \log Y)^{1.1}$$

and the last quantity is

$$o\left(\frac{Y}{(\log Y)^{1.1}}\right).$$

When b is scanned, every earlier multiple of p_2 has been deleted, since $b \leq 2X < F_X$. Thus no old accepted product is divisible by p_2 : if p_2 divided such a product, it would divide one old accepted factor, which would be an earlier deleted multiple of p_2 . Every new product containing b , on the other

hand, is divisible by p_2 . Hence no old-new collision can reject b . Old-old collisions are excluded by Lemma 3.2. Finally, two new products containing b would, after cancelling b , give equality between two final suffix products of the old accepted prefix; distinct final suffixes are strictly ordered by inclusion and each proper extension multiplies by an accepted term at least 2. Hence no new-new collision can reject b . Therefore each selected barrier is accepted. *Blocking.* Consider a surviving overlap-cancelled witness

$$\prod_{\text{old left}} d_i = n \prod_{\text{final suffix}} d_i$$

with the old-left block strictly before the final suffix. Suppose the old left block contains a selected barrier $b = p_1 p_2$ from shell X , and $n \leq N \leq F_X$. Then p_2 divides the left product, hence divides the right product. The corresponding terminal factor is either n or a suffix factor. It is not b , because the right side lies strictly to the right of the old block. It is not any selected barrier $c = r_1 r_2 \leq F_X$: if $p_2 = r_2$, global label uniqueness is violated; if $p_2 = r_1$, then

$$p_2 \leq (\log c)^{1.1} \leq (\log F_X)^{1.1} = o(X/(\log X)^{1.1}),$$

contradicting $p_2 \gg X/(\log X)^{1.1}$. Thus the terminal factor is a nonbarrier multiple of p_2 below F_X , hence belongs to D , which is impossible for either n or an accepted suffix factor.

Previous barriers and localization. Every sufficiently large nondeleted integer y which is not itself a selected barrier has a previous selected barrier $q_B(y) < y$ with

$$y - q_B(y) = O((\log y)^\beta).$$

If y lies after the selected barrier in its matched block, use that barrier. If y lies before it, then y is in the nonbarrier prefix of that matched block. Since $y \notin D$, this block cannot be the first matched block after a deleted run or after the shell boundary, because those prefixes were deleted. The preceding full block is therefore undeleted and matched, and its barrier is below y within two L_X -blocks.

Let n be a surviving nonempty rejection. It is not itself a selected barrier, since selected barriers are accepted. The final suffix in its chosen witness contains no selected barrier: if it contained $b = p_1 p_2$, then every old-left factor would be $< b \leq 2X_b < F_{X_b}$, where X_b is the shell of b . An old-left factor divisible by p_2 could not be a selected barrier, by global label uniqueness and the small-factor bound already used above; it would therefore be a deleted

nonbarrier multiple of p_2 . Thus the left product would not be divisible by p_2 , while the right product would be, a contradiction. If the final suffix began at or before $q_B(n)$, it would contain the accepted selected barrier $q_B(n)$. Consequently the suffix starts after $q_B(n)$, and its length is $O((\log n)^\beta)$.

Finally choose, for large N , the dyadic $X_0 = X_0(N)$ with

$$F_{X_0} < N \leq F_{2X_0}.$$

If the old-left maximum x in the chosen witness satisfies $x \leq 5X_0$, then n is $5X_0$ -smooth, because n divides the old-left product. Moreover

$$\frac{\log N}{\log(5X_0)} = (1 + o(1))\sqrt{\log \log N}.$$

Input 2.2 shows that these endpoints are $o(N)$. For all remaining endpoints, $x > 5X_0$. If x is a selected barrier, its horizon is at least N : the dyadic scale of x is above $2X_0$, and $X \mapsto F_X = X^{\lfloor \sqrt{\log \log X} \rfloor}$ is eventually increasing on dyadic scales, so this horizon is at least $F_{2X_0} \geq N$. The blocking argument forbids survival. If x is not a selected barrier, the previous barrier below x lies within $O((\log x)^\beta)$ and, for large N , above $2X_0$, since $x \leq N \leq F_{2X_0}$ makes every fixed power of $\log x$ equal to $o(X_0)$. The same monotonicity gives horizon at least N . The old-left block cannot contain that previous barrier, so it must start after it. Thus the old-left span is $O((\log n)^\beta)$. $\square$

5 Localized Nonempty Witnesses

Fix the deletion set and selected barriers from Proposition 4.1, and for a nonempty rejection n use the deterministic overlap-cancelled witness of Section 3. What remains is to turn the qualitative barrier information into a concrete numerical product identity, one in which every local shift is bounded by a fixed power of $\log X$.

Proposition 5.1 (Localized witness). *There is a fixed exponent $B > 20$ such that, in every sufficiently large dyadic shell $X < n \leq 2X$, all but $o(X)$ nonempty rejection endpoints have a chosen witness*

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j), \quad q = q_B(n), \quad n = q + h,$$

with the following properties:

$$\begin{aligned}
2 &\leq h \leq (\log X)^7, & 0 &\leq a_i < T_X, & 1 &\leq c_j \leq h, & T_X &= (\log X)^B, \\
b &= r + 1, & \ell &= b + s, & s &\geq 1, & r &\leq (\log X)^7, \\
\ell &\leq \lceil ((\log X)^7 + 1) \log_2(2X) \rceil + 2, \\
b &\geq c \frac{\log X}{\log \log X} \quad \text{for a fixed } c > 0, \\
x &= q^{b/\ell} + O(T_X) \quad \text{with an absolute implicit constant,} \\
(\log X)^{-B} &\leq 1 - \frac{b}{\ell} \leq B \frac{\log \log X}{\log X}, \\
x &< q < 2X.
\end{aligned}$$

The left factors $x - a_i$ are distinct accepted terms, the terminal values $q + c_j$ are distinct, and they are exactly $n = q + h$ together with the actual final length- r suffix before n .

Proof. The witness split gives an equality between an old block and n times a final suffix. The old block cannot be wholly contained in that suffix, since then its product would be at most the suffix product, whereas the right side is n times the suffix product with $n > 1$. If it overlaps the suffix, cancel the common contiguous overlap. Since the suffix is final, these are the only possibilities besides the old block already lying strictly before the suffix. If cancellation exhausted the suffix, then n would be an old block product and hence an empty-suffix rejection. Thus the chosen witness has the form

$$\prod_{t=u}^v d_t = n \prod_{t=c}^m d_t, \quad v < c \leq m.$$

Let $x = d_v$, let $w = d_u$, and let $r = m - c + 1$. Proposition 4.1, applied with ambient bound $2X$, removes $o(X)$ deterministic, inverse-horizon smooth, and finite-initial endpoints and gives

$$\begin{aligned}
2 &\leq h = n - q_B(n) \leq (\log X)^7, \\
r &\leq (\log X)^7, \\
x - w &\leq (\log X)^7, \\
x &> (\log X)^7.
\end{aligned}$$

After this removal, the remaining endpoints have $x > 5X_0(2X)$. This inverse-horizon scale satisfies $\log X_0(2X) = (1 + o(1)) \log X / \sqrt{\log \log X}$, so $X_0(2X) >$

$(\log X)^C$ for every fixed C , once X is large; in particular $x > (\log X)^7$. The lower bound $h \geq 2$ follows because the nonempty suffix starts after $q_B(n)$ and contains an accepted term $< n$.

The terminal side has $r + 1$ factors, each at most $2X$, while every old-left factor is an accepted integer at least 2. Hence

$$2^\ell \leq (2X)^{r+1},$$

which gives the stated upper bound for ℓ . Also $\ell \geq r + 2$: all terminal factors are larger than all old-left factors, so if $\ell \leq r + 1$, the terminal product would strictly exceed the old product.

Two elementary tails are negligible. First suppose

$$\frac{\ell}{r+1} \geq 1 + 60 \frac{\log \log X}{\log X}.$$

Since the old-left span is at most $(\log X)^7$,

$$(x - (\log X)^7)^\ell \leq (2X)^{r+1},$$

so $x \leq (\log X)^7 + X/(\log X)^{40} \leq X/(\log X)^{39}$ for large X . For fixed x, ℓ, r , the old-left block and its product are fixed, and the right side

$$n \cdot (\text{last } r \text{ accepted predecessors before } n)$$

is strictly increasing by Lemma 3.8. Thus there is at most one endpoint for each fixed x, ℓ, r . Here x has $O(X/(\log X)^{39})$ possible values, while ℓ and r have only fixed-polylogarithmically many possible values; the total is therefore $O(X/(\log X)^{20}) = o(X)$. Second, if

$$r \leq \left\lfloor \frac{\log X}{60 \log \log X} \right\rfloor,$$

then the same argument gives

$$x \leq (\log X)^7 + (2X)^{(r+1)/(r+2)} \leq \frac{3X}{(\log X)^{50}},$$

and the same fixed-product and monotonicity argument, now with only polylogarithmically many (ℓ, r) and $O(X/(\log X)^{50})$ possible values of x , gives $o(X)$ endpoints.

For the remaining endpoints, with $b = r + 1$,

$$b \geq c \frac{\log X}{\log \log X}, \quad \ell = b + s, \quad 1 \leq s < 60b \frac{\log \log X}{\log X}.$$

Choose fixed $B > 20$ so large that $T_X = (\log X)^B$ dominates all fixed polylogarithmic constants in the preceding estimates. The witness becomes

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j),$$

where $0 \leq a_i < T_X$, $1 \leq c_j \leq h$, and $n = q + h$ is included among the terminal values.

The terminal product lies between q^b and $(q + h)^b$, and the old product lies between $(x - (\log X)^7)^\ell$ and x^ℓ . Thus

$$q^{b/\ell} \leq x \leq (\log X)^7 + (q + h)^{b/\ell}.$$

Since $0 < b/\ell < 1$, the function $t^{b/\ell}$ is increasing and concave on $[q, q + h]$, with derivative at most 1. Hence $(q + h)^{b/\ell} - q^{b/\ell} \leq h$, and therefore

$$0 \leq x - q^{b/\ell} \leq (\log X)^7 + h \leq T_X$$

for large X , after enlarging B if necessary. The lower bound

$$1 - \frac{b}{\ell} \geq (\log X)^{-B}$$

follows from $s \geq 1$ and $\ell \leq (\log X)^B$; the upper bound follows from $s < 60b \log \log X / \log X$, after enlarging B .

Finally let $\theta = b/\ell$ and $\Delta = 1 - \theta \geq (\log X)^{-B}$. Since $q \geq X - (\log X)^7$, the function $\Delta \mapsto 1 - e^{-\Delta \log q}$ is increasing, and $(\log X)^{-B} \log q = o(1)$, we have

$$q - q^\theta = q(1 - e^{-\Delta \log q}) \gg \frac{X \log X}{(\log X)^B} \gg T_X.$$

Together with $0 \leq x - q^\theta \leq T_X$, this gives $x < q$. The inequality $q < 2X$ is immediate from $q < n \leq 2X$. $\square$

6 Terminal Support

A single product identity does not by itself bound the number of endpoints; what it yields is arithmetic information. Any large prime dividing the left-hand side of the localized identity must also divide one of the terminal factors on the right. The two lemmas of this section render this observation quantitative, extracting from each identity a family of independent large-prime congruences, which serve as the input to the counting argument of Section 7.

Lemma 6.1 (Smooth accepted left factors). *Let*

$$Y_X = \exp\left(\frac{\log X}{(\log \log X)^2}\right).$$

Among the localized endpoints of Proposition 5.1, those for which some accepted left factor $x - a_i$ has $P^+(x - a_i) \leq Y_X$ form an $o(X)$ family in the shell $(X, 2X]$.

Proof. By Lemma 3.7, the number of Y_X -smooth integers $\leq 2X$ is

$$o\left(\frac{X}{(\log X)^M}\right)$$

for every fixed M . Given such an integer $m = x - a_i$, there are at most T_X choices for x . For fixed x and ℓ there is at most one actual left block, and for each fixed suffix length r at most one endpoint. For the fixed left block has a fixed product P , while any endpoint counted with that block and this r must satisfy

$$P = n \prod_{\substack{a < n \\ a \text{ accepted} \\ a \text{ among last } r}} a,$$

and the right side is strictly increasing in n by Lemma 3.8. These at most T_X choices of x already account for the possible position of the smooth factor inside the left block. The remaining choices of ℓ and r are at most $O((\log X)^8)$ and $O((\log X)^7)$, respectively. Thus the endpoint count is

$$o\left(\frac{X}{(\log X)^M}\right) O((\log X)^{B+15}),$$

and taking M larger than this fixed loss gives $o(X)$. $\square$

Lemma 6.2 (Terminal-support extraction). *Consider a localized endpoint from Proposition 5.1, not counted in Lemma 6.1, and take X large enough that $(\log X)^7 < T_X$ and $T_X + (\log X)^7 < Y_X$. For every left factor choose*

$$p_i = P^+(x - a_i) > Y_X.$$

Put

$$\Lambda_X = \left\lceil \frac{\log(3X)}{\log Y_X} \right\rceil + 2.$$

Then the endpoint supplies at least $\lfloor \ell / \Lambda_X^2 \rfloor$ distinct local slab points (a_i, r_i) , $r_i = a_i + c_i$, with distinct r_i 's and carrying distinct prime labels $p_i > Y_X$ such that

$$p_i \mid x - a_i, \quad p_i \mid D_0 + r_i, \quad D_0 = q - x > 0.$$

Proof. Since p_i divides the left product, it divides one terminal value $q + c_i$, $1 \leq c_i \leq h$. This terminal value is unique, because two terminal values differ by less than $h < Y_X < p_i$. A label p_i appears on at most one left factor and at most one terminal value, since left factors differ by less than $T_X < Y_X$ and terminal values differ by at most $h < Y_X$.

A fixed terminal value $\leq n \leq 2X < 3X$ is incident to at most Λ_X chosen labels, since these labels are distinct primes and the value cannot have more than $\log(3X)/\log Y_X + 1$ distinct prime factors exceeding Y_X . Therefore the incidence graph from ℓ left vertices to terminal vertices contains a matching of size at least $\lfloor \ell/\Lambda_X \rfloor$: the ℓ left edges occupy at least $\lceil \ell/\Lambda_X \rceil$ terminal vertices, and choosing one incident edge over each occupied terminal gives a matching, since each left vertex has degree one. The matched labels are distinct.

For a matched edge, $p_i \mid x - a_i$ and $p_i \mid q + c_i$, so

$$p_i \mid (q + c_i) - (x - a_i) = D_0 + a_i + c_i.$$

Set $r_i = a_i + c_i$. For a fixed value ρ , all matched labels with $r_i = \rho$ divide the positive integer $D_0 + \rho$, where positivity follows from $D_0 > 0$ and $\rho = a_i + c_i \geq 1$. Since $D_0 + \rho \leq q - x + T_X + h < 3X$ for large X , there are at most Λ_X of them. Selecting one representative for each distinct r_i leaves at least $\lfloor \ell/\Lambda_X^2 \rfloor$ slab points: if $M \geq \lfloor \ell/\Lambda_X \rfloor$ matched edges remain and at most Λ_X have the same r_i , then $\lceil M/\Lambda_X \rceil \geq \lfloor \ell/\Lambda_X^2 \rfloor$: writing $\ell = a\Lambda_X^2 + r$ with $0 \leq r < \Lambda_X^2$, one has $\lfloor \ell/\Lambda_X \rfloor \geq a\Lambda_X$, and hence $\lceil \lfloor \ell/\Lambda_X \rfloor / \Lambda_X \rceil \geq a = \lfloor \ell/\Lambda_X^2 \rfloor$. The retained points have distinct labels and the stated divisibilities. $\square$

7 The Shifted-CRT Slab Count

Once the terminal support is available, the remaining count is carried out by a single proposition, whose proof rests on the determinant method. One interpolates a polynomial of low degree through the local points; the congruences accumulated above then force it to vanish at the global point; and a bound for the number of global points lying on a fixed curve of low degree completes the count. The argument divides into three parts: the interpolation step, an estimate for the nonlinear components of the auxiliary polynomial by means of Input 2.3, and a separate treatment of its line components, for which the determinant method is unavailable and the curvature of q^θ is used in its place.

Lemma 7.1 (Auxiliary interpolation). *Let Z be a set of $K \geq 1$ distinct integer points in a fixed polylogarithmic box*

$$|A|, |R| \leq (\log X)^{B_0},$$

and assume $K \log \log X = o(\log X)$. With

$$m = \left\lceil 4\sqrt{K} \right\rceil,$$

there is a nonzero nonconstant polynomial $G(A, R) \in \mathbb{Z}[A, R]$, of total degree at most m and height $\exp(O(m \log \log X))$, vanishing on Z . It may be chosen primitive, with a fixed sign convention for its first nonzero coefficient.

Proof. The number N_m of monomials $A^u R^v$, $u + v \leq m$, is $(m+1)(m+2)/2 > 2K$. The $K \times N_m$ evaluation matrix on Z has all entries bounded by $\exp(O(m \log \log X))$. Compare coefficient vectors in $[0, Q)^{N_m}$ under the K evaluation forms. For

$$Q \geq (3N_m H_0)^{K/(N_m-K)} + 1, \quad H_0 = \exp(O(m \log \log X)),$$

two vectors have the same image; their nonzero difference gives an integer kernel vector of height $\exp(O(m \log \log X))$. The corresponding polynomial vanishes on Z , and is not a nonzero constant because $K \geq 1$. Dividing its coefficient vector by the gcd of the coefficients and fixing the sign of the first nonzero coefficient preserves all these bounds and gives the primitive normalized choice. $\square$

Lemma 7.2 (Line-factor height). *Let $G(A, R) \in \mathbb{Z}[A, R]$ be primitive, nonzero, of total degree at most m , with*

$$m^2 = o(\log X), \quad H(G) = \exp(o(\log X)).$$

If

$$L(A, R) = \alpha A + \beta R + \gamma$$

is a primitive integer line factor of G over $\mathbb{Q}$, then

$$\max(|\alpha|, |\beta|, |\gamma|) = X^{o(1)}.$$

Proof. Homogenize G as a degree- m form; the line becomes the primitive homogeneous factor

$$L^h = \alpha A + \beta R + \gamma W.$$

Permute variables so the largest coefficient is the coefficient c of the homogenizing variable, and write the line as $aU + bV + cW$. Dehomogenizing at $W = 1$ gives a polynomial $P(U, V)$ of degree at most m , height $\leq H(G)$, divisible by $aU + bV + c$.

If $\max(|a|, |b|, |c|) = 1$ there is nothing to prove. Otherwise choose an integer slope $0 \leq t \leq m + 1$ such that $a + bt \neq 0$ and the cofactor in

$$P = (aU + bV + c)Q$$

does not vanish identically on $(U, V) = (Z, tZ)$. In this case $(a, b) \neq (0, 0)$. Such a slope exists because at most $m - 1$ slopes kill the nonzero cofactor and at most one slope makes $a + bt = 0$: writing $Q = \sum_j Q_j$ into homogeneous parts, if $Q(Z, tZ)$ is identically zero, then $Q_j(1, t) = 0$ for every j , and one fixed nonzero Q_j has degree at most $m - 1$. Then

$$P_t(Z) = P(Z, tZ)$$

is a nonzero one-variable integer polynomial of degree at most m , divisible by $(a + bt)Z + c$, and

$$\|P_t\|_2 \leq \exp(O(m \log m))H(G).$$

Mignotte's inequality, Input 2.4, gives

$$\max(|a + bt|, |c|) \leq 2^m \|P_t\|_2 \leq \exp(O(m \log m))H(G).$$

Since $|c|$ was the largest line coefficient and $m \log m = O(m^2) = o(\log X)$, the result follows. $\square$

Lemma 7.3 (Nonabsolute factors). *Let $H(A, R) \in \mathbb{Z}[A, R]$ be primitive, squarefree, $\mathbb{Q}$ -irreducible, not absolutely irreducible, and of total degree $d \geq 2$. Then*

$$\#\{P \in \mathbb{Q}^2 : H(P) = 0\} \leq d^2.$$

Proof. Over $\overline{\mathbb{Q}}$, the polynomial H is a product of at least two distinct Galois-conjugate absolute factors. A rational zero of H lies on one absolute factor, hence by Galois conjugacy lies on every conjugate factor. Choose two distinct factors. They have no common component, so Bezout bounds their common affine zero set by the product of their degrees, at most d^2 . $\square$

Proposition 7.4 (Shifted-CRT slab count). *Fix constants $A, B, B_1, \kappa > 0$. In a dyadic shell $X < n \leq 2X$, suppose the endpoints under consideration have integral localized data*

$$n = q + h, \quad 1 \leq h \leq (\log X)^A, \quad T_X = (\log X)^B,$$

$$x = q^\theta + O(T_X), \quad D_0 = q - x > 0, \quad 0 < \theta < 1,$$

where the implicit constant in $O(T_X)$ is uniform over all endpoints with the fixed parameter tuple.

$$(\log X)^{-B} \leq 1 - \theta \leq B \frac{\log \log X}{\log X}.$$

Assume that, for every fixed $\varepsilon > 0$, the number of parameter tuples, including h, θ , and K , is $O_\varepsilon(X^\varepsilon)$. Let

$$Y_X = \exp(\log X / (\log \log X)^\kappa).$$

Suppose each endpoint carries K distinct local slab points

$$z_i = (a_i, r_i), \quad 0 \leq a_i < T_X, \quad |r_i| \leq (\log X)^{B_1},$$

with distinct prime labels $p_i > Y_X$ satisfying

$$p_i \mid x - a_i, \quad p_i \mid D_0 + r_i.$$

If

$$K \log \log X = o(\log X), \quad \frac{K \log Y_X}{\sqrt{K} \log X} \rightarrow \infty,$$

then the number of such endpoints in $(X, 2X]$ is $o(X)$.

Proof. The proof proceeds in the three parts outlined above.

Auxiliary curves. Fix one localized parameter tuple and one retained K -set Z of local points. The number of possible Z 's is $X^{o(1)}$, since the local box is polylogarithmic and

$$\binom{(\log X)^{O(1)}}{K} \leq \exp(O(K \log \log X)) = X^{o(1)}.$$

Fix a constant C_G , depending only on the local box exponents B and B_1 , for which Lemma 7.1 supplies a primitive nonzero polynomial $G(A, R)$ of degree at most $m = \lceil 4\sqrt{K} \rceil$, height at most $\exp(C_G m \log \log X)$, and vanishing on Z . Among all such polynomials with first nonzero coefficient positive, choose the lexicographically first coefficient vector. Thus G depends only on the fixed parameters and the coordinate set Z , not on the endpoint-specific prime labels.

For every endpoint and every retained point,

$$(x, -D_0) \equiv (a_i, r_i) \pmod{p_i}.$$

Therefore each p_i divides $G(x, -D_0)$, and the distinctness of the labels gives

$$\prod_i p_i \mid G(x, -D_0).$$

If $G(x, -D_0) \neq 0$, then, since G has $O(m^2) = O(K)$ monomials and $|x|, |D_0| = O(X)$,

$$\log |G(x, -D_0)| = O(m \log X) = O(\sqrt{K} \log X).$$

But

$$\log \prod_i p_i \geq K \log Y_X,$$

and the ratio $K \log Y_X / (\sqrt{K} \log X)$ tends to infinity. This is incompatible with the preceding bound for large X . Hence $G(x, -D_0) = 0$.

Nonlinear components. Thus all endpoints lie on $X^{o(1)}$ auxiliary curves $G(A, R) = 0$, each of degree $m = X^{o(1)}$. Remove content and repeated factors, and then consider the primitive squarefree factorization over $\mathbb{Q}$. This does not change the zero set relevant to us: every point with $G(x, -D_0) = 0$ lies on at least one primitive squarefree $\mathbb{Q}$ -irreducible factor. Primitive line factors are deferred to the next paragraph. Nonlinear absolutely irreducible factors of degree $d \geq 2$ are integral affine curves. For one such factor, Input 2.3, in the box of height $O(X)$, gives

$$O(d^3 X^{1/d} (\log X + d)) \leq X^{1/2+o(1)}$$

points. The constant in Input 2.3 depends only on the fixed ambient dimension, and the squarefree factors of one auxiliary curve have total degree at most $m = X^{o(1)}$; summing this bound over the absolutely irreducible nonlinear factors still gives $X^{1/2+o(1)}$ points for that auxiliary curve. If a $\mathbb{Q}$ -irreducible nonlinear factor is not absolutely irreducible, Lemma 7.3 gives $O(d^2)$ rational points, so all nonabsolute factors of one auxiliary curve contribute only $X^{o(1)}$ points. Multiplication by the $X^{o(1)}$ curves still gives $o(X)$ endpoints, since the fixed parameter tuple includes h , and $(x, -D_0)$ determines $q = x + D_0$ and $n = q + h$.

Line components. It remains to count primitive line factors

$$\alpha A + \beta R + \gamma = 0.$$

Every such factor divides the original primitive auxiliary polynomial G over $\mathbb{Q}$, even after passing to the squarefree radical, so Lemma 7.2 applies to the same G . By Lemma 7.2,

$$M_X := \max(|\alpha|, |\beta|, |\gamma|) = X^{o(1)}$$

uniformly for all such factors. At the global point $R = -D_0 = x - q$,

$$(\alpha + \beta)x - \beta q + \gamma = 0. \quad (1)$$

Also $q \geq X/2$. Since $x = q^\theta + O(T_X)$ and

$$q^\theta \geq q/(\log X)^{O(B)} \gg T_X,$$

we have

$$x \gg q/(\log X)^{O(B)}.$$

For large X , neither β nor $\alpha + \beta$ can be zero in (1), otherwise x or q would be $X^{o(1)}$.

Define

$$F(q) = (\alpha + \beta)q^\theta - \beta q + \gamma.$$

Since $x = q^\theta + O(T_X)$, equation (1) gives, for a fixed constant C_1 ,

$$|F(q)| \leq C_1 M_X T_X. \quad (2)$$

On the sublevel set (2), we have

$$|F'(q)| \geq c|\beta|(1 - \theta). \quad (3)$$

To verify this, suppose $F'(q) = \delta$ with $|\delta| < |\beta|(1 - \theta)/4$; then

$$(\alpha + \beta)q^\theta = \frac{(\beta + \delta)q}{\theta}$$

and hence

$$F(q) = \frac{\beta q(1 - \theta)}{\theta} + \frac{\delta q}{\theta} + \gamma.$$

Using $|\beta| \geq 1$, $q \gg X$, $(1 - \theta) \geq (\log X)^{-B}$, and $\theta \geq 1/2$, the first term in this last display has magnitude $\gg X/(\log X)^B$. The δ -term has at most one quarter of that magnitude, and $|\gamma| \leq M_X = X^{o(1)}$ is negligible in comparison. Hence $|F(q)| \gg X/(\log X)^B$, which is much larger than $C_1 M_X T_X$. This contradicts (2), proving (3).

The derivative F' is monotone on $q > 0$, since $0 < \theta < 1$ and $\alpha + \beta \neq 0$, so F' has at most one zero. Thus F is monotone on at most two intervals, and the sublevel set $|F(q)| \leq C_1 M_X T_X$ has at most two interval components. Each component has length

$$O\left(\frac{C_1 M_X T_X}{|\beta|(1 - \theta)}\right) = X^{o(1)}$$

by the mean-value theorem and (3). Hence each line factor contributes $X^{o(1)}$ integer values of q , and, since h is fixed by the parameter tuple, $X^{o(1)}$ endpoints $n = q + h$. There are only $X^{o(1)}$ curves and line factors. Thus all line components contribute $o(X)$. Combining the nonlinear and line estimates proves the proposition. $\square$

8 Counting Nonempty Rejections

The pieces are now in place. Every exceptional family has already been shown to have shell size $o(X)$, and each endpoint that survives them carries enough terminal support to be counted by Proposition 7.4.

Proposition 8.1. *For the modified greedy construction using the deterministic deletion set from Proposition 4.1, the number of nonempty rejections $n \leq N$ is $o(N)$.*

Proof. Work in a dyadic shell $X < n \leq 2X$. Remove the deterministic deletions, empty-suffix endpoints, and the exceptional families from Propositions 4.1 and 5.1. These contribute $o(X)$ in the shell: the deterministic deletions are density zero, the empty-suffix endpoints are controlled by Corollary 3.6 with ambient bound $2X$, and the localization exceptions were proved either as shell $o(X)$ families or as global $o(N)$ families applied with $N = 2X$. These estimates were established independently of the present proposition.

For every remaining endpoint, Proposition 5.1 applies. Discard the $o(X)$ smooth-left family of Lemma 6.1. Then every left factor has a chosen label $p_i = P^+(x - a_i) > Y_X$, where

$$Y_X = \exp(\log X / (\log \log X)^2).$$

By Lemma 6.2, the endpoint supplies at least $\lfloor \ell / \Lambda_X^2 \rfloor$ slab points, where

$$\Lambda_X = \left\lceil \frac{\log(3X)}{\log Y_X} \right\rceil + 2 = (\log \log X)^2 + O(1).$$

Since Proposition 5.1 gives

$$\ell \geq b \geq c \frac{\log X}{\log \log X},$$

we may retain the lexicographically first

$$K_{\text{slab}} = \left\lfloor \frac{c \log X}{100(\log \log X)^5} \right\rfloor$$

of these extracted slab points. The factor 100 leaves room for the floor loss: for large X , $\Lambda_X \leq 2(\log \log X)^2$, so

$$\frac{\ell}{\Lambda_X^2} \geq \frac{c \log X}{4(\log \log X)^5},$$

and hence

$$\frac{\ell}{\Lambda_X^2} - 1 \geq \frac{c \log X}{100(\log \log X)^5} \geq K_{\text{slab}}.$$

Thus $\lfloor \ell/\Lambda_X^2 \rfloor \geq K_{\text{slab}}$.

The hypotheses of Proposition 7.4 hold with $A = 7$, $\kappa = 2$, $B_1 = B + 1$, $K = K_{\text{slab}}$, and $\theta = b/\ell$. Here Proposition 5.1 supplies $n = q+h$, $h \leq (\log X)^7$, $x = q^\theta + O(T_X)$ with a fixed implicit constant, $D_0 = q - x > 0$, $0 < \theta < 1$, and the two required bounds on $1 - \theta$. Lemma 6.2 gives the distinct slab points and distinct prime labels with the two divisibilities; retaining the first K_{slab} points preserves these properties. Finally,

$$|r_i| = |a_i + c_i| \leq T_X + (\log X)^7 \leq (\log X)^{B+1}$$

for large X . The localized parameter tuple is determined by h , (b, ℓ) , and K_{slab} . Since $h \leq (\log X)^7$ and, by Proposition 5.1,

$$1 \leq b, \ell \leq \lceil ((\log X)^7 + 1) \log_2(2X) \rceil + 2 = O((\log X)^8),$$

this gives only $O((\log X)^{23}) = X^{o(1)}$ possibilities. Moreover

$$K_{\text{slab}} \log \log X = o(\log X),$$

and, since

$$K_{\text{slab}} \geq \frac{c \log X}{200(\log \log X)^5}$$

for all large X ,

$$\frac{K_{\text{slab}} \log Y_X}{\sqrt{K_{\text{slab}}} \log X} = \frac{\sqrt{K_{\text{slab}}}}{(\log \log X)^2} \rightarrow \infty.$$

Thus Proposition 7.4 counts the surviving endpoints by $o(X)$. Adding back the finitely many $o(X)$ families removed above, every dyadic shell contains $o(X)$ nonempty rejections.

Finally sum dyadic shells. If f_j is the ratio of the number of nonempty rejections in $(2^j, 2^{j+1}]$ to 2^j , then $f_j \rightarrow 0$. For every $\varepsilon > 0$, choose J_0 so that $f_j \leq \varepsilon$ for $j \geq J_0$. The finite initial shells contribute $O(2^{J_0})$, while the dyadic tail up to N contributes at most

$$\varepsilon \sum_{2^j \leq N} 2^j \leq 2\varepsilon N.$$

Letting $\varepsilon \rightarrow 0$ gives $o(N)$. □

9 Proof of the Main Theorem

Proof of Theorem 1.1. Use the deterministic deletion set D from Proposition 4.1 and run the modified greedy construction. By Proposition 4.1, forced deletions contribute $o(N)$ rejected values. By Corollary 3.6, empty-suffix collision rejections contribute $o(N)$. By Proposition 8.1, nonempty collision rejections contribute $o(N)$.

Every rejected n belongs to the union of these three families. If $n \in D$, it is forced. If $n \notin D$, Lemma 3.3 gives an old-new witness $P = nS$. If some witness has $S = 1$, n is empty-suffix; otherwise the deterministic chosen witness has $S > 1$, and n is counted by the nonempty family. Disjointness is unnecessary. Hence the total rejected set has density zero, so the accepted set has density one. Lemma 3.2 gives distinct consecutive-block products at every finite stage. The increasing enumeration of the accepted set proves the theorem. $\square$

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